

Neha R

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SUMMARY

Final-year Integrated M.Tech (AI/ML) student at VIT Bhopal. Built full-stack production systems end-to-end — a race-condition-safe inventory reservation engine (Next.js + PostgreSQL + TypeScript), an agentic RAG pipeline (LangGraph + Pinecone), and a multimodal OCR pipeline — deployed with FastAPI and Docker. AWS Solutions Architect certified.

TECHNICAL SKILLS

Backend & APIs: TypeScript, Python, Next.js App Router, FastAPI (Async/Await), REST API Design, Node.js / Nest.js
Databases & Caching: PostgreSQL, MySQL, Prisma ORM, Redis Caching, Pessimistic Row Locking, Database Indexing
DevOps & Infrastructure: AWS (Solutions Architect Associate), Docker Containerization, Vercel, Neon, Git Version Control
Frontend & Validation: JavaScript, Web Content Scripts, DOM Manipulation, Zod Validation, TailwindCSS

PROJECTS

Allo Allocation Hub — Inventory Reservation Engine | May 2026

Next.js, TypeScript, PostgreSQL (Neon), Prisma, Zod

- Built a timed inventory reservation system solving checkout race conditions: holds units for 10 minutes at checkout, confirms on payment success, and auto-releases on expiry to eliminate double-selling without deflating active stock.
- Guaranteed concurrency correctness using PostgreSQL pessimistic row locking (SELECT FOR UPDATE inside an atomic transaction), ensuring exactly one 201 and one 409 response status code when concurrent requests race for the final available unit.
- Implemented lazy expiry cleanup on every catalog read (indexed on status and expiresAt) and engineered a responsive frontend countdown timer mirroring the server-side verification state with live UI synchronization.
- Enforced system idempotency on the reservation endpoint via an Idempotency-Key header mapped to a unique-indexed database column, returning the original cached response on network retries without repeating side effects.

Agentic AI eBook Assistant | Feb 2026

LangGraph, Python, FastAPI, Pinecone

- Built a state-machine based RAG system using LangGraph and reduced out-of-context responses on a 50-query eval set from 30% to under 5% using relevance scoring.
- Engineered an asynchronous Ingestion Pipeline that processes PDF data into 512-dimensional vectors stored in a Pinecone index for high-speed semantic retrieval.
- Developed a Multi-Node Logic Flow (Retriever, Relevance Checker, Generator, Validator) to provide deterministic, traceable agent behavior and real-time Gradio interaction.
- Achieved <2s Response Times by leveraging FastAPI's non-blocking I/O operations and optimized vector similarity search.

AutoForm AI — Multimodal Document Processing | Nov 2025

Python, JavaScript, FastAPI, spaCy NLP, OpenCV

- Built an offline data engineering pipeline combining image preprocessing and transformer-based recognition, reducing processing latency by 78%.
- Eliminated manual data labelling by building an intelligent field classification system using spaCy NLP and regex patterns, automatically tagging extracted text into Name, Phone, Email, and ID categories with zero user input.
- Built a self-learning rule engine in the content script that silently maps form field labels to Vault keys on every manual input, persisting rules in Chrome local storage to skip fuzzy matching on future autofills.

EXPERIENCE

Salesforce Intern – Impaqtive

Oct 2025 – Dec 2025

- Built Screen and Record-Triggered Automation Flows, configured custom objects and data schemas, and wrote SOQL queries for data retrieval across CRM modules.
- Explored Salesforce's AI layer (Einstein/Agentforce) and its agent-building paradigm — directly informed my approach to designing multi-agent AI systems with tool-use and state management in Python.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal

Aug 2022 – May 2027

Integrated M.Tech in Artificial Intelligence & Machine Learning

CGPA: 8.69 / 10.0

CERTIFICATIONS

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| AWS Certified Solutions Architect – Associate Verified Badge | 2026 |
| Salesforce Developer with Agentblazer Champion | 2025 |
| Cloud Computing – NPTEL | 2024 |
| Applied Machine Learning in Python – University of Michigan | 2023 |